

FIRST & FOLLOW Sets — Advanced Practice

Compilers · Lecture 4: Parsers · Top-down / predictive parsing

Name: _____ Group: _____ Date: _____

Part	Focus	Exercises	Points	Time
A	Nullable symbols, long-range propagation	1, 2	22	30 min
B	Fixed-point iteration, cyclic dependencies	3	12	15 min
C	Equivalent grammars (left recursion, left factoring)	4, 5	22	25 min
D	Error analysis and reasoning about the definitions	6, 7	16	20 min
E	From FIRST/FOLLOW to LL(1) decisions	8	16	20 min
F	Design challenge (reverse engineering)	9	12	10 min
	Total		100	120 min

Instructions. Write final answers in the tables provided and use set notation, e.g. $\{a, b, \epsilon\}$, writing $\emptyset$ for the empty set. Show intermediate work (nullable sets, partial sets, iteration steps) in the working space: partial credit is awarded for correct reasoning. The time shown next to each exercise is a guide; if you are stuck, move on and come back.

Conventions used throughout

- ϵ is the empty string and $\$$ the end-of-input marker. ϵ may belong to a FIRST set but **never** to a FOLLOW set; $\$$ belongs to FOLLOW of the start symbol.
- Upper-case letters are non-terminals; lower-case letters and keywords are terminals. The left-hand side of the first production is the start symbol.
- A symbol (or string) is **nullable** if it can derive ϵ . For a string $\alpha = X_1X_2\dots X_n$, $\text{FIRST}(\alpha)$ contains $\text{FIRST}(X_1) \setminus \{\epsilon\}$; it also contains $\text{FIRST}(X_2) \setminus \{\epsilon\}$ if X_1 is nullable, and so on; and $\epsilon \in \text{FIRST}(\alpha)$ iff every X_i is nullable. $\text{FIRST}(\epsilon) = \{\epsilon\}$.
- Left recursion and ϵ -productions are allowed unless an exercise says otherwise: FIRST and FOLLOW are defined for *any* context-free grammar.

30 min

8 pts · ~10 min

$$C \rightarrow c \mid d \mid \varepsilon$$

- | | FIRST | FOLLOW |
|---|-------|--------|
| S | | |
| A | | |
| C | | |

- ### Working space

Exercise 2 — Nullable prefixes, nullable suffixes

14 pts · ~20 min

Grammar G_2

$$S \rightarrow A B C$$
$$A \rightarrow a \mid B \mid \epsilon$$
$$B \rightarrow b \mid C y \mid \varepsilon$$
$$C \rightarrow c \ C \mid z$$

(a) [8 pts] Compute FIRST and FOLLOW for every non-terminal of G_2 .

	FIRST	FOLLOW
S		
A		
B		
C		

(b) [3 pts] Compute the following FIRST sets of *strings* (grammar symbols mixed freely):

$$\text{FIRST}(B \times A) = \text{FIRST}(A \cup B) = \text{FIRST}(C \cup B) =$$

(c) [3 pts] In the start production $S \rightarrow A B C$, the symbol A is followed by the nullable symbol B , yet $\$ \notin \text{FOLLOW}(A)$. Explain why, and say what would have to change in the grammar for $\$$ to enter $\text{FOLLOW}(A)$.

Working space

Exercise 3 — Chasing FOLLOW around a cycle

12 pts · ~15 min

Grammar G_3 $S \rightarrow A p \mid q B r \mid C s$ $A \rightarrow x B$ $B \rightarrow y C \mid \epsilon$ $C \rightarrow z A \mid w$

(a) [3 pts] Give the nullable non-terminals and the FIRST set of every non-terminal.

	FIRST
S	
A	
B	
C	

(b) [6 pts] Compute the FOLLOW sets with the *iterative* algorithm and record the state after each pass. In every pass, scan the productions **top to bottom** and, inside a production, the non-terminal occurrences **left to right**; update sets **in place** (a change made earlier in a pass is visible to the later steps of the same pass). Stop after the first pass in which nothing changes — that last pass must still be recorded. Leave unused columns empty.

	Init.	Pass 1	Pass 2	Pass 3	Pass 4
FOLLOW(S)					
FOLLOW(A)					
FOLLOW(B)					
FOLLOW(C)					

(c) [3 pts] Draw the *inclusion graph* with a node per non-terminal and an edge $X \rightarrow Y$ whenever a production forces $\text{FOLLOW}(X) \subseteq \text{FOLLOW}(Y)$. What structure do you observe, and what does it prove about the three sets, independently of the number of passes?

Working space

Exercise 4 — Eliminating indirect left recursion

12 pts · ~15 min

The grammar G_4 below is indirectly left-recursive ($S \Rightarrow Aa \Rightarrow Sda$). Applying the standard elimination algorithm (order S, A) produces the equivalent grammar G_4' . The primed non-terminal is a new symbol.

Grammar G_4 (left-recursive)

$$S \rightarrow Aa \mid b$$

$$A \rightarrow Ac \mid Sd \mid \epsilon$$

Grammar G_4' (after elimination)

$$S \rightarrow Aa \mid b$$

$$A \rightarrow bdA' \mid A'$$

$$A' \rightarrow cA' \mid adA' \mid \epsilon$$

- (a) [3 pts] Compute FIRST and FOLLOW of S and A **directly on G_4** (the left recursion does not prevent this).

	FIRST	FOLLOW
S		
A		

- (b) [4 pts] Compute FIRST and FOLLOW of every non-terminal of G_4' .

	FIRST	FOLLOW
S		
A		
A'		

- (c) [2 pts] Both grammars generate the same language. Which of the sets in (a) and (b) coincide and which differ? Explain the difference in one or two sentences.
- (d) [3 pts] Using *only* FIRST and FOLLOW, decide whether G_4' is LL(1). If not, name a non-terminal, the two productions in conflict, and the terminal on which they collide.

Working space

Grammar G_5

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S → if E then S
  | if E then S else S
  | while E do S
  | begin L end
  | a
L → S ; L | S
E → id | num

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Terminals: if then else while do begin end ; a id num.

(a) [2 pts] Left-factor G_5 (call the new non-terminals S' and L') and write the resulting grammar.

New grammar

(b) [6 pts] Compute FIRST and FOLLOW for every non-terminal of the left-factored grammar.

	FIRST	FOLLOW
S		
S'		
L		
L'		
E		

(c) [2 pts] The left-factored grammar is still not LL(1). Identify the offending production pair, the terminal involved, and say which of the two FOLLOW-related conditions is violated.

Working space

Part D — Error analysis and reasoning

20 min

Exercise 6 — Mark the student's work

10 pts · ~12 min

Grammar G_6

$S \rightarrow A B e \mid C f$

$A \rightarrow a A \mid \epsilon$

$B \rightarrow b B \mid C A \mid \epsilon$

$C \rightarrow c C \mid d$

A student produced the table below. **Exactly five of the eight entries are wrong.** For each wrong entry, (i) circle it, (ii) name the underlying mistake in a few words, and (iii) write the correct set in the last column.

	FIRST	FOLLOW	Correct set(s) and mistake
S	{a, b, c, d}	{\$}	
A	{a, ϵ }	{b, c, d, e, \$}	
B	{b, c, d}	{e, ϵ }	
C	{c, d}	{a, f}	

Scoring: 2 pts per wrong entry (identification + correct set + mistake). No credit is lost for leaving correct entries unmarked, but marking a correct entry as wrong costs 1 pt.

Exercise 7 — True or false? Justify

6 pts · ~8 min

Decide whether each statement is true for **every** context-free grammar. Justify in one or two lines; for a false statement, give a small counter-example grammar (1 pt each).

1.	Some non-terminal A of some grammar has $\epsilon \in \text{FOLLOW}(A)$.	T / F	
2.	If $A \rightarrow \alpha B \beta$ is a production and $\beta \Rightarrow^* \epsilon$, then $\text{FOLLOW}(A) \subseteq \text{FOLLOW}(B)$.	T / F	
3.	If $\text{FIRST}(\alpha) \cap \text{FIRST}(\beta) = \emptyset$ for every pair of alternatives $A \rightarrow \alpha \mid \beta$, then the grammar is LL(1).	T / F	
4.	FIRST and FOLLOW sets cannot be computed for a left-recursive grammar.	T / F	
5.	If $\text{FIRST}(A) = \text{FIRST}(B)$ for two non-terminals A and B, then $\text{FOLLOW}(A) = \text{FOLLOW}(B)$.	T / F	
6.	$\text{FOLLOW}(S) = \{\$ \}$ for the start symbol S of every grammar.	T / F	

Exercise 8 — Table construction and conflict diagnosis

16 pts · ~20 min

Grammar G_8 $S \rightarrow A B \mid C e$ $A \rightarrow a A \mid B c \mid \epsilon$ $B \rightarrow b B \mid C$ $C \rightarrow d C \mid \epsilon$ **(a)** [6 pts] Compute FIRST and FOLLOW for every non-terminal.

	FIRST	FOLLOW
S		
A		
B		
C		

(b) [5 pts] Build the LL(1) parsing table. Place each production $X \rightarrow \alpha$ under every terminal of $\text{FIRST}(\alpha) \setminus \{\epsilon\}$, and, when α is nullable, under every terminal of $\text{FOLLOW}(X)$ (including \$). Write several productions in a cell if they collide.

	a	b	c	d	e	\$
S						
A						
B						
C						

(c) [3 pts] Circle every multiply-defined cell. For each, state whether it is a FIRST/FIRST conflict or a FIRST/FOLLOW conflict, and which nullable derivation causes it.**(d)** [2 pts] Give two input strings that both begin with the same terminal but require *different* productions for S, so that a single lookahead symbol cannot decide.

Working space

10 min

12 pts · ~10 min

- $\text{FIRST}(S) = \{a, c, \epsilon\}$, $\text{FIRST}(A) = \{a, \epsilon\}$, $\text{FIRST}(B) = \{c, \epsilon\}$;
- $\text{FOLLOW}(S) = \{b, \$\}$, $\text{FOLLOW}(A) = \{b, c, \$\}$, $\text{FOLLOW}(B) = \{b, \$\}$;
- the grammar is LL(1).

Your grammar

	FIRST	FOLLOW
S		
A		
B		

LL(1) check

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